

SESSION 12 • TUTORIAL 3

Tutorial 3: SFEE & Unsteady Flow — Unit I Finale

Six fully solved problems, multi-device chains, CAT-I revision map + practice set with answers

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Purpose of this tutorial. The Unit-I finale: consolidated problem practice on the SFEE and unsteady flow, a mixed multi-device problem, and a complete revision map of Unit I for CAT-I (which covers Units I & II). Six fully solved problems and a practice set are provided.

1. Equations on One Line

$$q - w = \Delta h + \Delta KE/2000\text{-form} + \Delta PE \quad | \quad V_2 = \sqrt{(2000\Delta h + V_1^2)} \quad | \quad u_2 = h_{\text{line}} \quad | \quad T_2 = \gamma T_{\text{line}}$$

2. Worked Problems

WORKED EXAMPLE T3.1 — Nozzle exit velocity

Steam enters a nozzle with negligible velocity at $h_1 = 3000$ kJ/kg and leaves at $h_2 = 2762$ kJ/kg (adiabatic, no work). Find the exit velocity.

Solution.

- $V_2 = \sqrt{(2000(h_1 - h_2))} = \sqrt{(2000 \times 238)}$.
- $V_2 = \sqrt{476000} = 690$ m/s.

ANSWER $V_2 \approx 690$ m/s

Note: The 2000 converts kJ/kg to m^2/s^2 .

WORKED EXAMPLE T3.2 — Compressor power with heat loss

Air flows at 0.8 kg/s through a compressor; h rises from 300 to 510 kJ/kg; heat lost to cooling water is 40 kJ/kg; ΔKE , ΔPE negligible. Find the power required.

Solution.

- $w = q - \Delta h = (-40) - 210 = -250$ kJ/kg.
- Power input = $0.8 \times 250 = 200$ kW.

ANSWER $w = -250$ kJ/kg | 200 kW input

WORKED EXAMPLE T3.3 — Charging a bottle

An insulated, rigid, evacuated 0.05 m^3 bottle is filled from an air line at 500 kPa, 290 K ($R = 0.287$, $\gamma = 1.4$). Find T_2 and the mass admitted.

Solution.

- $T_2 = \gamma T_{\text{line}} = 406$ K.
- $m = p_2 V / (RT_2) = (500 \times 0.05) / (0.287 \times 406) = 0.2146$ kg.

ANSWER $T_2 = 406$ K | $m \approx 0.215$ kg

WORKED EXAMPLE T3.4 — Boiler heat duty

Feed water enters a boiler at $h_1 = 505$ kJ/kg and leaves as steam at $h_2 = 2960$ kJ/kg at 1.2 kg/s (no work; ΔKE , ΔPE negligible). Find the rate of heat supply.

Solution.

1. $q = h_2 - h_1 = 2455$ kJ/kg.
2. $\dot{Q} = 1.2 \times 2455 = 2946$ kW ≈ 2.95 MW.

ANSWER $\dot{Q} \approx 2.95$ MW

WORKED EXAMPLE T3.5 — Boiler + turbine in series (multi-device)

The steam from the boiler of Problem T3.4 ($h = 2960$ kJ/kg, 1.2 kg/s) expands adiabatically in a turbine to $h = 2510$ kJ/kg with negligible ΔKE . Find the turbine power and the fraction of the boiler heat input converted to work.

Solution.

1. Turbine: $w = h_1 - h_2 = 2960 - 2510 = 450$ kJ/kg.
2. Power = $1.2 \times 450 = 540$ kW.
3. Fraction = $540/2946 = 0.183 \rightarrow 18.3\%$ (a foretaste of cycle efficiency in Unit II).

ANSWER Power = 540 kW | $\approx 18.3\%$ of boiler heat

Note: Chaining devices = chaining SFEEs; the exit state of one is the inlet state of the next.

WORKED EXAMPLE T3.6 — Turbine supplied through a throttle (mixed concepts)

Steam at $h = 3050$ kJ/kg is throttled to a lower pressure before entering a turbine, which it leaves at $h = 2600$ kJ/kg (turbine adiabatic, ΔKE negligible). Find the turbine work per kg and comment on the throttle's effect.

Solution.

1. Throttle: h unchanged $\Rightarrow$ turbine inlet enthalpy = 3050 kJ/kg.
2. $w = 3050 - 2600 = 450$ kJ/kg.
3. The throttle wastes pressure (availability) without changing h ; with the throttle removed the same exit pressure would yield MORE work — quantified by the second law in Unit II.

ANSWER $w = 450$ kJ/kg

3. Unit-I Revision Map for CAT-I (Units I & II)

Topic block	Expected question type	Marks
Systems, properties, point/path functions	Definitions, 2-mark, MCQ	2–4
Quasi-static process, equilibrium	Short explanation with sketch	2–4
Displacement work and p–V diagram	Numerical: polytropic / isothermal / multi-step	8–13
Zeroth law and temperature scales	Statement + conversion / new-scale numerical	2–6
First law — closed systems	Numerical: process or cycle table	8–13
SFEE: derivation and devices	Derivation (8) or turbine/nozzle/compressor numerical	8–13
Unsteady flow (charging/discharging)	Short derivation or direct numerical	4–8

KEY POINT Revision priority: first-law numericals $\rightarrow$ SFEE devices $\rightarrow$ work integrals $\rightarrow$ unsteady flow $\rightarrow$ theory definitions. Practise the cycle-table method and the SFEE derivation until they are automatic.

4. Practice Problems (answers given)

#	Problem	Answer
P1	Steam ($h_1 = 2900$ kJ/kg) leaves a nozzle at 500 m/s from rest. Find h_2 .	2775 kJ/kg
P2	A pump raises water ($v = 0.001$ m ³ /kg) from 100 kPa to 2 MPa adiabatically with negligible ΔKE . Estimate w per kg ($\approx v\Delta p$).	≈ -1.9 kJ/kg
P3	An insulated evacuated vessel is filled from a line at 310 K ($\gamma = 1.4$). Find T_2 .	434 K

5. Unit-I Closing Summary

- The unit's storyline: define the system $\rightarrow$ demand equilibrium $\rightarrow$ idealise processes (quasi-static) $\rightarrow$ quantify work ($\int p \, dV$) $\rightarrow$ establish temperature (zeroth law) $\rightarrow$ conserve energy (first law, closed then open) $\rightarrow$ extend to transients.
- Master equations: $Q - W = \Delta U$; $q - w = \Delta h + \Delta KE + \Delta PE$; $u_2 = h_{line}$.
- Unit II asks the question the first law cannot answer: in which direction, and how well, can energy be converted? — heat engines, the second law and entropy.

Text: P.K. Nag, "Engineering Thermodynamics", 6th Edn., Tata McGraw-Hill — Ch. 5 (First Law Applied to Flow Processes). Reference: Cengel & Boles, "Thermodynamics: An Engineering Approach", 8th Edn.